Exercise on Coin Flipping

- How many ways are there to repeatedly flip a coin until the number of tails is more than the number of heads or the coin is flipped for five times? *Example 1*: The first flip is head, the second flip is tail and the third flip is tail. No fourth flip. *Example 2*: All the five flips are head.
- Solution:
- There are 2^5 =32 bit-strings of length 5. Let 1 denote tails.
- If the first flip is a tail you stop the experiment – i.e., you exclude 2^4 bit-strings that start with 1.
- The only other outcome where you stop the experiment before 5 flips is when the first 3 flips are HTT (i.e., 011). In this case, you exclude 01 100, 01 101, 01 110, 01 111. Thus, the total number of ways is: $2^5-2^4-4=12$.
- Add the two cases of early termination (1 and 011), the total is 12+2=14 ways.
- Specifically, the possible strings are: 1, 011, 00 000, 00 001, 00 010, 00 011, 00 100, 00 101, 00 110, 00 111, 01 000, 01 001, 01 010, 01 011.